gain in turbulence modeling, so was seen as an acceptable trade off, especially since the values for $y^+ \approx 1$ or lower in the vast majority of the mesh.

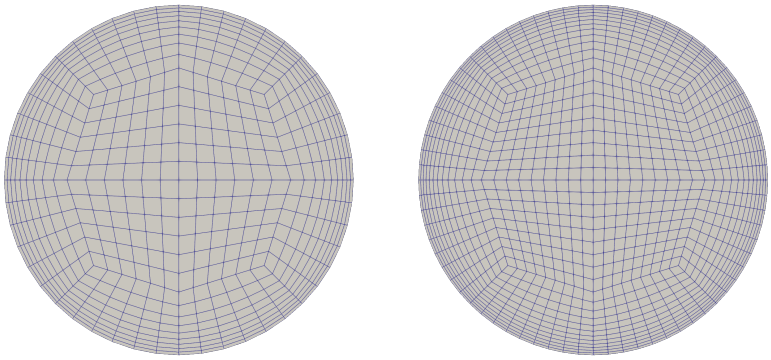


Figure 6.4: Loop Pipe Cross Sections

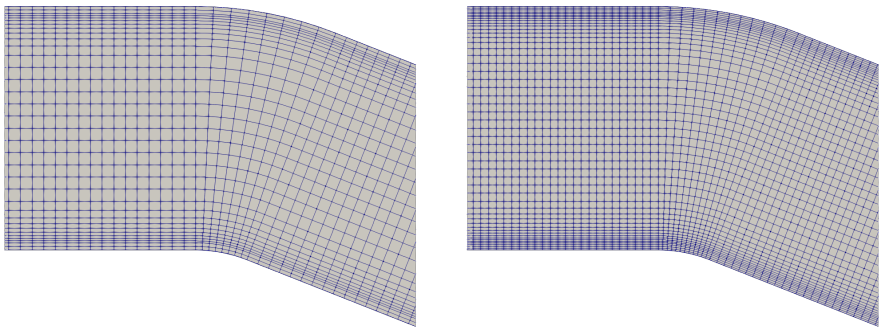


Figure 6.5: Loop Pipe Bend

With that bound set, we can compute the numerical uncertainty of the simulation results before finally determining the validation uncertain for each of the QoIs studied. The full matrix including the uncertainty for turbulent cases can be found in Appendix C. This is true for each table of uncertainties unless explicitly stated otherwise. Below, are the values for numerical uncertainty calculated from the two-mesh GCI. $\mathrm{TP}_{\mathrm{mean}}$ and $\mathrm{TW}_{\mathrm{mean}}$ relate to the average numerical uncertainty among the TP and TW series of probes of probes respectively. $\dot{m}_{\mathrm{flow}}$ is the numerical uncertainty associated with the cross-sectional mass flux through the